JAVA ASSIGNMENT-5

Q. 1) Difference between default , parameterized and copy constructors ?

Answer :

Constructor :

A constructor in java is a special member that is called when an object is created.

Types :

1) Default Constructor :
   When you do not create any constructor, java automatically creates one.

e.g.
```
class ABC {
        ABC () {
                System.out.println ("Default constructor");
        }
    public static void main(String[] args) {
                ABC  a = new ABC();
    }
}
```

2) Paramaterized Constructor :
   Constructor with parameters is called parameterized constructor.

e.g.
```
class Student {
        int id;
        String name;
```

```
        Student (int i ,  String n)  {

                This.id =  i ;

                This.name  = n;

        }

public static void main(String[] args)   {

                Student s = new Student (101, “Rahul”);

                System.out.println (s.id+” “+s.name);

}

}
```

3) Copy Constructors :

It is used to create a new object by copying the the values of an existing object of the same class. Java does not have inbuilt copy Constructor. We have to create it manually.
The primary purpose is to enable the creation of independent duplicates of objects, ensuring that changes to the copied object do not affect the original.

e.g.

```
class Student   {

        int id ;

        String name;

        //  original constructor

        Student (int i , String n)  {
                this.id = i ;

                this.name = n;

        }

        // Copy constructor


        Student (Student s)  {

                id = s.id;
```

```
        name = s.name;
    }
}
```

Q. 2) **What is the use of "this" keyword ?**

Answer :

"this" keyword is a reference variable that refers to the current object of a class.

Uses :

1) Differntiate local and instance variable.
2) Refer current class instance variable.
3) Call current class method.
4) Call current class constructor.
5) Improve code clarity and readability.
6) It cannot be used in static method.

Q. 3) **What is the use of "super" keyword ?**

Answer :

super :

super is a reference variable that refers to the immediate parent class object and is used to access parent class members.

Uses :

1) Access parent class instance variable.
2) Call parent class methods.
3) Call parent class constructor.

Rules :

1) Refers only to the immediate parent class.
2) Cannot be used in static context.
3) Automatically called when if not written explicitly.
4) Used with inheritance only.

Q. 4) What is a "static" keyword used for ?

Answer :

Static keyword is used to create class-level members that belong to the class itself , not to individual objects.

Uses :

1) Cannot use "this" or "super" keywords.
2) Accessed using "className.member"
3) Used to create class level variables shared by all objects.
4) Used to define methods that can be called without creating object.
5) Used to save memory by sharing common data.
6) Used for initializing static data using static block.
7) Used to create static nested class.
8) Used for utility methods like math, arrays, collections..etc

Q. 5) What are the static blocks and static methods ?

Answer :

Static Blocks **:**

A static block is a block of code declared using the "static" keyword that belongs to the class and is executed automatically when the class is loaded into JVM memory.

Memory allocated in Method area or metaspace. After loading class it executed (before main method). It access only static members. It cannot use 'super' and 'this' keywords. In one class we can use multiple static blocks and it executed in top to bottom order.

e.g.
```
class Demo {

  static {

    System.out.println("Static block executed");

  }


  public static void main(String[] args) {

    System.out.println("Main method");

  }
```

}

Static Methods :

A static method is a method declared using the 'static' keyword that belongs to the class rather than to any object and can be called without creating an instance. Static method can be called using className.method(). For calling a static method no need of object reference. It exists in a memory once per class.

Uses :

1) Perform common operation .
2) Create utility / helper methods.
3) Used in math , array , collection classess.

Rules :

1) Static method can access only static members.
2) Cannot use this or super keywords.
3) Can be overloaded.
4) Cannot be overridden.(only method hiding).
5) Memory allocated in method area / metaspace.

Q. 6) Write a program to find sum of n natural numbers.

Answer :

```
class Test {

  static void show() {

    System.out.println("Static method");

  }


  public static void main(String[] args) {

    Test.show();   // preferred way

    show();       // allowed inside same class

  }

}
```

Output :

Enter a target no.(last no) :

```
10
Sum of first 10 numbers is : 55
```

Q. 7) Write a program to reverse a String.

Answer :

```java
	import java.util.Scanner;

public class ReverseString {

	public static void main(String[] args) {
		// TODO Auto-generated method stub

		Scanner sc = new Scanner(System.in);

		System.out.println("Enter a string : ");
		String str = sc.nextLine();

		char[] ch = str.toCharArray();
		int n = ch.length;



		for (int i = 0; i < n/2; i++) {
			char temp = ch[i];
			ch[i] = ch[n-1-i];
			ch[n-1-i] = temp;
		}
```

```java
        String reverse_str = new String(ch);

        System.out.println("Reverse Array is : "+reverse_str);

        sc.close();

    }

}
```

Output :

```
    Enter a string :
    Ajinkya Pathak
    Reverse Array is : kahtaP ayknijA
```

Q. 8) Write a program to check if a String is palindrome.

Answer :

```java
import java.util.Scanner;

public class StringPalindrome {

    public static void main(String[] args) {
        // TODO Auto-generated method stub

        Scanner sc = new Scanner(System.in);

        System.out.println("Enter a String : ");
        String str = sc.next();
```

```
		char[] ch = str.toCharArray();
		int n = ch.length;

		for (int i = 0; i < n/2; i++) {
			char temp = ch[i];
			ch[i] = ch[n-1-i];
			ch[n-1-i] = temp;
		}

		String str2 = new String(ch);

		if (str.equalsIgnoreCase(str2))
			System.out.println("String is Palindrome.");
		else
			System.out.println("String is not Palindrome.");

		sc.close();
	}
}
```

Output :

Enter a String :

level

String is Palindrome.

Q. 9) Write a program to count vowels and consonants in a String.

Answer :

```
import java.util.Scanner;
```

```java
public class CountVowelsAndConsonants {

	public static void main(String[] args) {
		// TODO Auto-generated method stub

		Scanner sc = new Scanner(System.in);

		System.out.println("Enter a String : ");
		String str = sc.nextLine();

		char[] ch = str.toCharArray();
		int vowel = 0;
		int conso = 0;

		for (int i = 0; i < ch.length; i++) {

			if (ch[i]=='a' || ch[i]=='e' || ch[i]=='i' || ch[i]=='o' || ch[i]=='u' )
				vowel++;
			else
				conso++;
		}

		System.out.println("Total Vowels : "+vowel);
		System.out.println("Total Consonants : "+conso);

		sc.close();

	}
```

}

Output :

Enter a String :

Information Technology

Total Vowels : 7

Total Consonants : 15

Q. 10) Write a program to count words in a sentence.

Answer :

```
import java.util.Scanner;

public class CountCharacters {

	public static void main(String[] args) {
		// TODO Auto-generated method stub

		Scanner sc = new Scanner(System.in);

		System.out.println("Enter a string : ");
		String str = sc.nextLine();

		char[] ch = str.toCharArray();

		int count=0;

		for (int i = 0; i < ch.length; i++) {
			if (ch[i] != ' ')
```

```
                count++;
          }
          System.out.println("Total Characters : "+count);

     }

}
```

Output :

Enter a string :

Information Tech

Total Characters : 15